UCC KYC GATEWAY | VENDOR INTEGRATION

KYC Verification API

Vendor Integration Documentation

Version 2.0.0 · November 2026 · Base URL: http://10.33.33.30:8001

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KYC Verification API — Vendor Integration Documentation

Version: 2.0.0
Date: November 2026
Vendor-Facing Base URL: `http://10.33.33.30:8001`
Authentication: `X-API-Key` header (vendor-assigned key)

1. Overview

The KYC Verification API enables Mobile Network Operators (MNOs) and vendors to submit SIM registration information for validation and compliance. The gateway acts as a secure reverse proxy between your application and UCC:

- You connect only to `http://10.33.33.30:8001`
- The gateway validates your API key locally
- The gateway injects all provider credentials (NIRA partner keys, certificates, signing logic) internally
- Requests are forwarded to UCC; responses are returned to you unchanged

Important: You must NOT connect directly to `192.168.5.18`. Only use `http://10.33.33.30:8001`. The UCC upstream address is internal.

Network Flow

```
Vendor Application → http://10.33.33.30:8001 (KYC Gateway) → UCC (https://192.168.5.18:443)
```

2. Authentication

Every business endpoint requires an API key supplied in the `X-API-Key` request header. A missing or unrecognised key returns HTTP 401 with `error_code: UNAUTHORIZED`. The key identifies your MNO/vendor and is scoped to your data.

Request Header

```
X-API-Key: <your-assigned-vendor-key>
Content-Type: application/json
```

- Keys are issued by the gateway administrator via a separate secure channel
- Do not share keys, commit them to source control, or include them in URLs
- `GET /health`, `/docs`, `/redoc`, and `/openapi.json` do not require authentication

3. Identity Providers and Biometric Formats

The `id_type` field selects the authority that verifies the subscriber. Each provider requires a specific biometric format:

ID Type	Provider	Status	Biometric Format	Freshness Window
NATIONAL ID	NIRA	Live	MOSIP_SBI_SIGNED (device-signed block)	~30 seconds
REFUGEE ID	UNHCR	Live	WSQ or JPEG (fingerprint image)	Not time-critical
PASSPORT	NCIC	Planned	Not applicable	N/A
COI	URSB	Planned	None (organisation has no biometric)	N/A
REGISTRY_PENDING	Undecided	Planned	None	N/A

NATIONAL ID — NIRA (MOSIP_SBI_SIGNED)

NIRA does not accept a bare template. It requires the block produced and signed by a MOSIP-certified SBI capture device. Submit that block exactly as the device produced it, inside `biometric.blocks`. UCC relays it to NIRA unmodified — it does not parse, validate, or reshape it. Altering the block after capture will fail signature verification at NIRA.

Two constraints: 1. The signed block must reach NIRA within about **30 seconds** of capture — capture immediately before calling `/identity_verification` 2. The block's signed `transactionId` must equal the `transaction_id` sent in the request

REFUGEE ID — UNHCR

Send a fingerprint image record in `biometric.data`, Base64 encoded. Accepted formats: `WSQ`, `JPEG`, `ISO_19794_4`, `ISO_19794_6`. If you send `WSQ`, the gateway automatically converts it to JPEG before forwarding to UCC — no client-side change is required.

Important Biometric Handling Rules

- Biometric data is processed transiently and is **never persisted** by this API
- Biometric data appears **only** on `POST /identity_verification` — never on a registration endpoint
- Application logs must never record biometric data
- Fingerprint is the only modality currently submitted; IRIS is reserved in the schema

---

4. Two-Step Registration Flow

Registration is a two-step operation: verification first, registration second.

```
STEP 1: POST /ownership_check          → ownership eligibility decision
STEP 2: POST /identity_verification    → identity + biometric verification, returns verification_reference
STEP 3: POST /individual_registrations → register SIM using verification_reference
```

`verification_reference` Rules

- Bound to the `id_type` + `id_number` verified in step 2; presenting it with a different identity returns `VERIFICATION_REFERENCE_MISMATCH`
- Valid for **24 hours**; after that, `VERIFICATION_REFERENCE_EXPIRED`
- Reusable within its lifetime — a failed SIM provisioning does not force the subscriber to present a finger again. Every use is recorded
- Required by every registration endpoint, including PUT updates and every record inside a batch

---

5. Endpoint Specifications

5.1 POST /ownership_check

Checks whether a subscriber is eligible to register a new SIM card. Call this **before** capturing a biometric.

Request

Field	Type	Required	Description
subscriber_type	string	Yes	"individual" or "corporate"
id_type	string	Yes (individual)	"NATIONAL ID" , "REFUGEE ID" , "PASSPORT" , "COI" , "REGISTRY_PENDING"
id_number	string	Yes (individual)	National ID number or refugee ID number

Field	Type	Required	Description
registration_number	string	Yes (corporate)	Company/business registration number

Request Example — Individual (National ID)

```
{
  "subscriber_type": "individual",
  "id_type": "NATIONAL ID",
  "id_number": "CF123456789ABCDE"
}
```

Request Example — Individual (Refugee ID)

```
{
  "subscriber_type": "individual",
  "id_type": "REFUGEE ID",
  "id_number": "WA9-00115526"
}
```

Request Example — Corporate

```
{
  "subscriber_type": "corporate",
  "id_type": "COI",
  "registration_number": "80020001234"
}
```

Response — Eligible

```
{
  "status": "success",
  "data": {
    "allowed": true,
    "reason_code": "LIMIT_NOT_EXCEEDED",
    "message": "Subscriber can register SIM",
    "current_count": 2,
    "limit": 5
  }
}
```

Response — Not Eligible

```
{
  "status": "success",
  "data": {
    "allowed": false,
    "reason_code": "LIMIT_EXCEEDED",
    "message": "Subscriber has exceeded SIM ownership limit",
    "current_count": 5,
    "limit": 5
  }
}
```

5.2 POST /identity_verification

Verifies a subscriber against the relevant identity registry (NIRA, UNHCR, URSB, NCIC) and returns a `verification_reference` for use at registration.

Important: A failure to match is **not** an HTTP error. It is HTTP 200 with `verified: false` and `reason_code: IDENTITY_REJECTED`. Branch on `data.verified` and `data.reason_code`, never on the HTTP status alone.

Request — NATIONAL ID (NIRA)

The middleware executes the full three-step NIRA flow internally (token → certificate → signed auth). You supply only identity and biometric blocks — no NIRA credentials, certificates, or cryptography are required from you.

Field	Type	Required	Description
id_type	string	Yes	"NATIONAL ID"
id_number	string	Yes	National ID number (e.g., CM702356039748)
first_name	string	No	Given names (e.g., KEVIN)
last_name	string	No	Surname (e.g., SSONKO)
date_of_birth	string	Yes	Date of birth in YYYY-MM-DD format
transaction_id	string	Yes	10-digit transaction ID from the signed biometric block
biometric	object	Yes	Signed MOSIP SBI block object
biometric.biometric_type	string	Yes	"FINGERPRINT"
biometric.format	string	Yes	"MOSIP_SBI_SIGNED"
biometric.blocks	array	Yes	Array of signed biometric blocks from MOSIP SBI capture

Request Example — NATIONAL ID

```
{
  "id_type": "NATIONAL ID",
  "id_number": "CM702356039748",
  "first_name": "KEVIN",
  "last_name": "SSONKO",
  "date_of_birth": "1970-11-06",
  "transaction_id": "1234567890",
  "biometric": {
    "biometric_type": "FINGERPRINT",
    "format": "MOSIP_SBI_SIGNED",
    "blocks": [
      {
        "specVersion": "0.9.5",
        "data": "eyJhbGciOiJSUzI1NiIsIng1YyI6WyJNSUlE...",
        "hash": "A1B2C3...",
        "sessionKey": "kR9dP...",
        "thumbprint": "3f5a9c..."
      }
    ]
  }
}
```

Request — REFUGEE ID (UNHCR)

Field	Type	Required	Description
id_type	string	Yes	"REFUGEE ID"
id_number	string	Yes	Refugee ID number (e.g., WA9-00115526)
first_name	string	Yes	First name (e.g., SAMSON)
last_name	string	Yes	Last name (e.g., LADU)
biometric	object	Yes	Fingerprint image object
biometric.biometric_type	string	Yes	"FINGERPRINT"
biometric.format	string	Yes	"WSQ" , "JPEG" , "ISO_19794_4" , or "ISO_19794_6"

Field	Type	Required	Description
biometric.data	string	Yes	Base64-encoded fingerprint image data

Request Example — REFUGEE ID (WSQ)

```
{
  "id_type": "REFUGEE ID",
  "id_number": "WA9-00115526",
  "first_name": "SAMSON",
  "last_name": "LADU",
  "biometric": {
    "biometric_type": "FINGERPRINT",
    "format": "WSQ",
    "data": "/6D/qAB6TklTVF9DT00gOQpQSVhfV0lEVEggMjk2..."
  }
}
```

Request — PASSPORT (NCIC — Not Yet Integrated)

```
{
  "id_type": "PASSPORT",
  "id_number": "P12345678",
  "visa_validity": "2027-06-30"
}
```

Returns `verified: false` with `reason_code: VERIFICATION_UNAVAILABLE`. A `verification_reference` is still issued so registration can proceed as unverified.

Response — Verified (NATIONAL ID)

```
{
  "status": "success",
  "data": {
    "verified": true,
    "reason_code": "IDENTITY_ACCEPTED",
    "message": "Identity verified via NIRA",
    "provider": "NIRA",
    "provider_status": "LIVE",
    "verification_reference": "vrf_FSMUJHVL3Z7QEMIZ",
    "verified_at": "2026-11-03T18:44:08.461Z",
    "expires_at": "2026-11-04T18:44:08.461Z",
    "matched_attributes": null
  }
}
```

Response Header: `X-Verification-Reference: vrf_FSMUJHVL3Z7QEMIZ`

Response — Verified (REFUGEE ID)

```
{
  "status": "success",
  "data": {
    "verified": true,
    "reason_code": "IDENTITY_ACCEPTED",
    "message": "Identity verified",
    "provider": "UNHCR",
    "provider_status": "LIVE",
    "verification_reference": "vrf_01JBK3RM9P2W6YD1",
    "verified_at": "2026-08-24T09:14:22.481Z",
    "expires_at": "2026-08-25T09:14:22.481Z",
    "matched_attributes": {
      "date_of_birth": "MATCH",
      "sex": "F"
    }
  }
}
```

Response — Not Verified


```
{
  "status": "success",
  "data": {
    "verified": false,
    "reason_code": "IDENTITY_REJECTED",
    "message": "Biometric did not match the identity record",
    "provider": "NIRA",
    "provider_status": "LIVE",
    "verification_reference": null,
    "verified_at": null,
    "expires_at": null,
    "matched_attributes": null
  }
}
```


Response — Provider Unavailable


```
{
  "status": "success",
  "data": {
    "verified": false,
    "reason_code": "VERIFICATION_UNAVAILABLE",
    "message": "No verification provider is currently available for PASSPORT",
    "provider": "NCIC",
    "provider_status": "PLANNED",
    "verification_reference": "vrf_01JBK4TN0Q3X7ZE2",
    "expires_at": "2026-08-25T09:14:22.481Z",
    "matched_attributes": null
  }
}
```


Validation Error Examples

Error Code	Message	Scenario
INVALID_BIOMETRIC	NATIONAL ID requires format MOSIP_SBI_SIGNED	Wrong biometric format for provider
MISSING_FIELD	transaction_id is required when format is MOSIP_SBI_SIGNED	Missing transaction_id for signed block
INVALID_ID	Invalid id_type	Unrecognised id_type value
MISSING_FIELD	biometric is required for this id_type	Biometric omitted for NATIONAL ID or REFUGEE ID

5.3 POST /individual_registrations

Registers a SIM for an individual subscriber. Call `POST /identity_verification` first and send the `verification_reference` it returned.

This endpoint does NOT carry a biometric and does NOT call a provider. Identity was already established in step 2, and the reference is the proof of it.

Request Fields

Field	Type	Required	Description	Example
verification_reference	string	Yes	Reference from identity verification	vrf_01JBK2QX7M4T8ZC3
msisdn	string	Yes	MSISDN in international format	256712345678
sim_registration_type	string	Yes	"NEW" , "SIM SWAP" , "RE_REGISTRATION"	NEW

Field	Type	Required	Description	Example
sim_type	string	Yes	"SIM" , "ESIM" , "ISIM"	SIM
sim_registration_datetime	string	Yes	ISO 8601 datetime	2025-11-01T10:22:00Z
id_type	string	Yes	"NATIONAL ID" , "REFUGEE ID" , "PASSPORT"	NATIONAL ID
id_number	string	Yes	ID number	CM702356039748
nationality	string	Yes	Subscriber nationality	UGANDAN
surname	string	Yes	Surname	KATO
given_name	string	Yes	Given names	JOHN
gender	string	Yes	"MALE" , "FEMALE" , "UNDEFINED"	MALE
date_of_birth	string	Yes	Date of birth (YYYY-MM-DD)	1995-05-12
id_validity	string	No	ID expiry date (YYYY-MM-DD)	2030-12-31
visa_permit_number	string	No	Visa/permit number	VISA12345
visa_validity	string	No	Visa expiry date (YYYY-MM-DD)	2027-06-30
agent_code	string	Yes	Agent code (uppercase alphanumeric)	AGT001
agent_number	string	Yes	Agent MSISDN	256712345678
sales_district	string	Yes	Sales district	KAMPALA
gps_coordinates	string	Yes	GPS coordinates (lat,lon)	32.5825,0.3476
activation_date	string	Yes	Activation date (YYYY-MM-DD)	2025-10-01

Request Example — NATIONAL ID

```
{
  "verification_reference": "vrf_FSMUJHVL3Z7QEMIZ",
  "msisdn": "256724072418",
  "sim_registration_type": "NEW",
  "sim_type": "SIM",
  "sim_registration_datetime": "2026-11-04T10:53:00Z",
  "id_type": "NATIONAL ID",
  "id_number": "CM702356039748",
  "nationality": "UGANDAN",
  "surname": "SSONKO",
  "given_name": "KEVIN",
  "gender": "MALE",
  "date_of_birth": "1970-11-06",
  "agent_code": "AGENT11",
  "agent_number": "256724072418",
  "sales_district": "KAMPALA",
  "gps_coordinates": "0.33147638683656766, 32.58604725173673",
  "activation_date": "2026-11-04",
  "id_validity": "2030-12-31"
}
```

Request Example — REFUGEE ID

```
{
  "verification_reference": "vrf_01JBK3RM9P2W6YD1",
```

```
  "msisdn": "256724184554",
  "sim_registration_type": "NEW",
  "sim_type": "SIM",
  "sim_registration_datetime": "2026-11-04T10:53:00Z",
  "id_type": "REFUGEE ID",
  "id_number": "WA9-00115526",
  "nationality": "SOUTH SUDAN",
  "surname": "LADU",
  "given_name": "SAMSON",
  "gender": "MALE",
  "date_of_birth": "2001-01-01",
  "agent_code": "AGENT11",
  "agent_number": "256724072418",
  "sales_district": "KAMPALA",
  "gps_coordinates": "0.33147638683656766, 32.58604725173673",
  "activation_date": "2026-11-04",
  "id_validity": "2027-07-28"
}
```

Response — Accepted

```
{
  "status": "accepted",
  "reason_code": "REGISTRATION_ACCEPTED",
  "message": "SIM registration stored"
}
```

Error Responses

Error Code	Message	Scenario
VERIFICATION_REQUIRED	A verification_reference from identity_verification is required	No reference supplied
VERIFICATION_REFERENCE_INVALID	Verification reference is not recognised	Reference not found
VERIFICATION_REFERENCE_EXPIRED	Verification reference expired; re-verify the subscriber	Reference older than 24 hours
VERIFICATION_REFERENCE_MISMATCH	Verification reference does not match the supplied identity	Reference bound to different identity
LIMIT_EXCEEDED	Subscriber has exceeded SIM ownership limit	Ownership limit reached at commit time
DUPLICATE_REQUEST	SIM already registered	MSISDN already has active registration
INVALID_MSISDN	Invalid MSISDN format	MSISDN doesn't match ^2567\d{8}$
INVALID_AGENT	Agent number format is invalid	Agent number invalid

5.4 PUT /individual_registrations/{msisdn}

Fully replaces an existing subscriber registration. This is a **full re-verification** event — a fresh `verification_reference` is required.

Path Parameter

Name	Type	Required	Description
msisdn	string	Yes	Subscriber MSISDN in international format

Request Body

Same fields as `POST /individual_registrations`, including a fresh `verification_reference`.

Request Example

```
{
```

```
  "verification_reference": "vrf_01JBK6VQ2S5Z9BG4",
  "msisdn": "256712345678",
  "sim_registration_type": "SIM SWAP",
  "sim_type": "SIM",
  "sim_registration_datetime": "2025-12-01T09:00:00Z",
  "id_type": "NATIONAL ID",
  "id_number": "CF123456789ABCDE",
  "nationality": "UGANDAN",
  "surname": "KATO",
  "given_name": "JOHN PAUL",
  "gender": "MALE",
  "date_of_birth": "1994-05-12",
  "id_validity": "2030-12-31",
  "agent_code": "AGT001",
  "agent_number": "256712345678",
  "sales_district": "KAMPALA",
  "gps_coordinates": "32.5825,0.3476",
  "activation_date": "2025-10-01"
}
```

Response — Updated

```
{
  "status": "updated",
  "reason_code": "REGISTRATION_ACCEPTED",
  "message": "Subscriber information successfully updated"
}
```

Note: To correct a detail without re-proving identity (e.g., surname change after marriage, typing error), use `PATCH /individual_registrations/{msisdn}` instead. That operation needs no verification and no biometric.

5.5 PATCH /individual_registrations/{msisdn}

Partial amendment of permitted fields on an existing registration. No verification reference or biometric is required.

Amendable Fields

Field	Type	Description
surname	string	Subscriber surname
given_name	string	Subscriber given names
gender	string	"MALE" , "FEMALE" , "UNDEFINED"
id_validity	string	ID expiry date (YYYY-MM-DD)
visa_validity	string	Visa expiry date (YYYY-MM-DD)
visa_permit_number	string	Visa/permit number

Request Example

```
{
  "change_reason": "LEGAL_NAME_CHANGE",
  "evidence_reference": "MC/2026/004412",
  "changes": {
    "surname": "MUKASA",
    "given_name": "GRACE"
  }
}
```

Response — Amended

```
{
  "status": "amended",
  "reason_code": "REGISTRATION_AMENDED",
  "message": "Subscriber details amended",
  "previous_values": {
    "surname": "AYO"
```

```
  }
}
```

5.6 POST /individual_registrations/{msisdn}/ownership_change

Transfers SIM ownership to a new subscriber.

Request Fields

Field	Type	Required	Description
verification_reference	string	Yes	Reference for the new owner
effective_date	string	Yes	Transfer effective date (YYYY-MM-DD)
new_owner	object	Yes	New owner identity details
new_owner.id_type	string	Yes	"NATIONAL ID" , "REFUGEE ID" , "PASSPORT"
new_owner.id_number	string	Yes	New owner ID number
new_owner.nationality	string	Yes	New owner nationality
new_owner.surname	string	Yes	New owner surname
new_owner.given_name	string	Yes	New owner given names
new_owner.gender	string	Yes	"MALE" , "FEMALE" , "UNDEFINED"
new_owner.date_of_birth	string	Yes	New owner date of birth (YYYY-MM-DD)
new_owner.id_validity	string	No	New owner ID expiry date
new_owner.visa_validity	string	No	New owner visa expiry date
previous_owner_consent	object	Yes	Evidence of previous owner consent
previous_owner_consent.method	string	Yes	"SUBSCRIBER_CONFIRMED" , "SIGNED_FORM" , "SUCCESSION" , "OPERATOR_ATTESTED"
previous_owner_consent.reference	string	No	Consent reference (e.g., USSD code)
previous_owner_consent.obtained_at	string	Yes	Consent obtained datetime (ISO 8601)
agent_code	string	Yes	Agent code (uppercase alphanumeric)
agent_number	string	Yes	Agent MSISDN

Request Example

```
{
  "verification_reference": "vrf_01JBK7WR3T6A0CH5",
  "effective_date": "2026-08-26",
  "new_owner": {
    "id_type": "NATIONAL ID",
    "id_number": "CM20000000001",
    "nationality": "UGANDAN",
    "surname": "OKELLO",
    "given_name": "PETER",
    "gender": "MALE",
    "date_of_birth": "1990-03-04",
    "id_validity": "2031-06-30"
  },
```

```
  "previous_owner_consent": {
    "method": "SUBSCRIBER_CONFIRMED",
    "reference": "USSD-8837721",
    "obtained_at": "2026-08-26T09:14:00Z"
  },
  "agent_code": "AGT001",
  "agent_number": "256712345678"
}
```

Response — Transferred

```
{
  "status": "transferred",
  "reason_code": "OWNERSHIP_CHANGED",
  "message": "SIM ownership transferred",
  "previous_owner": {
    "id_type": "NATIONAL ID",
    "id_number": "CF123456789ABCDE"
  },
  "effective_date": "2026-08-26"
}
```

5.7 POST /individual_registrations/{msisdn}/deregistration

Deregisters/ends a SIM registration.

Request Example

```
{
  "reason_code": "SUBSCRIBER_REQUEST",
  "effective_date": "2026-08-26",
  "agent_code": "AGT001",
  "agent_number": "256712345678",
  "notes": "Subscriber reported the handset as stolen"
}
```

Response — Deregistered

```
{
  "status": "deregistered",
  "reason_code": "REGISTRATION_DEREGISTERED",
  "message": "SIM registration ended",
  "effective_date": "2026-08-26",
  "released_count": 1
}
```

5.8 POST /corporate_registrations

Registers a SIM card for a corporate entity. Verify the organisation first through `POST /identity_verification`.

Note: The nominated representative is **recorded, not verified**. No identity or biometric check is performed on them.

Request Fields (subset)

Field	Type	Required	Description	Example
verification_reference	string	Yes	Reference for the organisation	vrf_01JBK8XS4U7B1DJ6
msisdn	string	Yes	MSISDN	256712345679
sim_type	string	Yes	"SIM" , "ESIM" , "ISIM"	SIM
sim_registration_type	string	Yes	"NEW" , "SIM SWAP"	NEW
sim_registration_datetime	string	Yes	ISO 8601 datetime	2025-11-01T10:22:00Z

Field	Type	Required	Description	Example
id_type	string	Yes	"COI" or "REGISTRY_PENDING"	COI
organisation_type	string	Yes	"COMPANY" , "BUSINESS" , etc.	COMPANY
organisation_name	string	Yes	Organisation name	ABC LIMITED
registration_number	string	Yes	Registration number	80020012345678
registration_validity	string	No	Registration expiry date	2028-12-31
registered_address	string	Yes	Registered address	KAMPALA ROAD
tax_identification_number	string	Yes	TIN	TIN123456
contact_number	string	Yes	Contact MSISDN	256712345679
representative	object	Yes	Representative details (recorded, not verified)	—
agent_code	string	Yes	Agent code	AGT001
agent_number	string	Yes	Agent MSISDN	256712345678
sales_district	string	Yes	Sales district	KAMPALA
gps_coordinates	string	Yes	GPS coordinates	32.5825,0.3476
activation_date	string	Yes	Activation date (YYYY-MM-DD)	2025-10-01

Request Example

```
{
  "verification_reference": "vrf_01JBK8XS4U7B1DJ6",
  "msisdn": "256712345679",
  "sim_type": "SIM",
  "sim_registration_type": "NEW",
  "sim_registration_datetime": "2025-11-01T10:22:00Z",
  "id_type": "COI",
  "organisation_type": "COMPANY",
  "organisation_name": "ABC LIMITED",
  "registration_number": "80020012345678",
  "registration_validity": "2028-12-31",
  "registered_address": "KAMPALA ROAD",
  "tax_identification_number": "TIN123456",
  "contact_number": "256712345679",
  "representative": {
    "id_type": "NATIONAL ID",
    "id_number": "CF123456789ABCDE",
    "nationality": "UGANDAN",
    "surname": "KATO",
    "given_name": "JOHN",
    "gender": "MALE",
    "date_of_birth": "1983-05-12",
    "validity": "2030-12-31"
  },
  "agent_code": "AGT001",
  "agent_number": "256712345678",
  "sales_district": "KAMPALA",
  "gps_coordinates": "32.5825,0.3476",
  "activation_date": "2025-10-01"
}
```

5.9 Batch Operations

Submit multiple registrations in one request. Include a unique `Idempotency-Key` header to prevent duplicates.

POST /individual_registrations/batch

```
{
  "records": [
    {
      "verification_reference": "vrf_01JBK2QX7M4T8ZC3",
      "msisdn": "256712345678",
      "sim_registration_type": "NEW",
      "sim_type": "SIM",
      "sim_registration_datetime": "2025-11-01T10:22:00Z",
      "id_type": "NATIONAL ID",
      "id_number": "CF123456789ABCDE",
      "nationality": "UGANDAN",
      "surname": "KATO",
      "given_name": "JOHN",
      "gender": "MALE",
      "date_of_birth": "1995-05-12",
      "agent_code": "AGT001",
      "agent_number": "256712345678",
      "sales_district": "KAMPALA",
      "gps_coordinates": "32.5825,0.3476",
      "activation_date": "2025-10-01"
    }
  ]
}
```

Response — Batch Accepted

```
{
  "status": "accepted",
  "job_id": "batch_123456",
  "message": "Batch accepted for processing"
}
```

GET /individual_registrations/batch/{job_id}

Poll batch status and results.

Response — Batch Status

```
{
  "job_id": "batch_123456",
  "status": "completed",
  "total_records": 100,
  "processed_records": 100,
  "successful_records": 95,
  "failed_records": 5,
  "results": [
    {
      "msisdn": "256712345678",
      "status": "accepted",
      "reason_code": "REGISTRATION_ACCEPTED",
      "verification_reference": "vrf_01JBK2QX7M4T8ZC3",
      "error_message": null
    },
    {
      "msisdn": "256712345679",
      "status": "rejected",
      "reason_code": "INVALID_MSISDN",
      "verification_reference": null,
      "error_message": "Invalid MSISDN format"
    }
  ]
}
```

5.10 GET /compliance_report

Retrieve a compliance report scoped to your API key.

Response

```
{
  "status": "success",
  "data": {
    "valid_records": 12000,
    "invalid_records": 340,
    "missing_information": 87
  }
}
```

5.11 GET /health

Health check (no authentication required).

Response

```
{
  "status": "ok",
  "service": "KYC Verification API",
  "timestamp": "2026-11-04T10:53:00.000+03:00"
}
```

---

6. Complete Workflow Examples

Example A: NATIONAL ID Registration

Step 1 — Ownership Check:

```
POST /ownership_check
X-API-Key: your-key
{"subscriber_type": "individual", "id_type": "NATIONAL ID", "id_number": "CM702356039748"}
```

→ `{"allowed": true, "reason_code": "LIMIT_NOT_EXCEEDED", ...}`

Step 2 — Identity Verification (NIRA flow handled by middleware):

```
POST /identity_verification
X-API-Key: your-key
{
  "id_type": "NATIONAL ID",
  "id_number": "CM702356039748",
  "first_name": "KEVIN",
  "last_name": "SSONKO",
  "date_of_birth": "1970-11-06",
  "transaction_id": "1234567890",
  "biometric": {
    "biometric_type": "FINGERPRINT",
    "format": "MOSIP_SBI_SIGNED",
    "blocks": [<signed MOSIP SBI block>]
  }
}
```

→ Response header: `X-Verification-Reference: vrf_FSMUJHVL3Z7QEMIZ` → Body: `{"verified": true, "reason_code": "IDENTITY_ACCEPTED", ...}`

Step 3 — Registration:

```
POST /individual_registrations
X-API-Key: your-key
{
  "verification_reference": "vrf_FSMUJHVL3Z7QEMIZ",
  "msisdn": "256724072418",
  ...
}
```

→ `{"status": "accepted", "reason_code": "REGISTRATION_ACCEPTED", ...}`

Example B: REFUGEE ID Registration

Step 2 — Identity Verification:

```
POST /identity_verification
X-API-Key: your-key
{
  "id_type": "REFUGEE ID",
  "id_number": "WA9-00115526",
  "first_name": "SAMSON",
  "last_name": "LADU",
  "biometric": {
    "biometric_type": "FINGERPRINT",
    "format": "WSQ",
    "data": "<base64 WSQ fingerprint>"
  }
}
```

→ Body: `{"verified": true, "reason_code": "IDENTITY_ACCEPTED", "verification_reference": "vrf_...", ...}`

Step 3 — Registration: Same as Example A, using the refugee verification reference.

7. Common HTTP Responses

HTTP Status	Error Code	Meaning
200 OK / 201 Created / 202 Accepted	—	Request successful. Always inspect the response body for business logic results.
401 Unauthorized	UNAUTHORIZED	Missing or invalid X-API-Key .
403 Forbidden	LIMIT_EXCEEDED	Subscriber has exceeded SIM ownership limit.
403 Forbidden	VERIFICATION_REFERENCE_INVALID	Invalid, expired, or already-used (replayed) verification reference. Perform identity verification again.
403 Forbidden	VERIFICATION_REFERENCE_EXPIRED	Reference is older than 24 hours.
403 Forbidden	VERIFICATION_REFERENCE_MISMATCH	Reference was issued for a different identity.
403 Forbidden	CONSENT_REQUIRED	previous_owner_consent is required for ownership change.
404 Not Found	NOT_FOUND	Subscriber record not found.
409 Conflict	DUPLICATE_REQUEST	SIM already registered or batch idempotency key reused.
422 Unprocessable Entity	MISSING_FIELD	Missing required field.
422 Unprocessable Entity	INVALID_FIELD	Invalid field value/format.
422 Unprocessable Entity	INVALID_BIOMETRIC	Biometric format does not match provider requirements.
502 Bad Gateway	UCC_UPSTREAM_UNAVAILABLE	Gateway could not reach UCC upstream service.
500 Internal Server Error	SYSTEM_ERROR	Unexpected system error.

8. What the Middleware Handles Internally (Not Your Concern)

The following are **middleware operator secrets** and are never shared with vendors:

Item	Description
NIRA partner private key & certificate	Used for JWS signing of /nira/auth
NIRA encryption certificate	Used for thumbprint + session key encryption
NIRA client ID, secret key, app ID	NIRA partner credentials
NIRA MISP license key, partner ID, partner API key	NIRA auth headers
Client-side cryptography	RSA-OAEP, AES-GCM, JWS RS256 implementation
UCC upstream URL	Internal routing (https://192.168.5.18)

Your integration surface is limited to the endpoints documented above with your `X-API-Key`.

9. Go-Live Checklist

- [ ] Confirm your server can reach `10.33.33.30:8001` (you do not need direct access to `192.168.5.18`)
- [ ] Obtain a unique `X-API-Key` from the middleware operator via a secure channel
- [ ] Review the OpenAPI contract at `http://10.33.33.30:8001/openapi.json`
- [ ] For NATIONAL ID: capture a fresh MOSIP SBI biometric block and send it within ~30 seconds of the auth call
- [ ] Use HTTPS via a trusted reverse proxy before using this connection outside a controlled private network

10. Interactive Documentation

When the gateway is running, interactive Swagger UI documentation is available at:

`http://10.33.33.30:8001/docs`

OpenAPI JSON schema:

`http://10.33.33.30:8001/openapi.json`